

Project: Hosting a Static Website on AWS S3

Introduction

This project demonstrates the deployment of a static website using Amazon Web Services (AWS) Simple Storage Service (S3). AWS S3 is a highly scalable, durable, and secure cloud storage service that can also be used for hosting static websites consisting of HTML, CSS, JavaScript, images, and other static assets.

The project involves creating an S3 bucket, configuring permissions, uploading website files, enabling static website hosting, and making the website publicly accessible through an AWS-generated endpoint URL.

The developed website is a portfolio website containing Home, About, Projects, and Contact pages. The Contact page is integrated with Google Sheets using Google Apps Script to store user inquiries.

Key Features

I) Cost-Effective Hosting

AWS S3 provides a low-cost hosting solution where users pay only for the storage and bandwidth they consume.

II) High Availability

Website content is stored redundantly across AWS infrastructure, ensuring high availability and reliability.

III) Scalability

The hosted website can handle increasing visitor traffic without requiring manual server management.

IV) Easy Deployment

Website files can be uploaded directly to the S3 bucket without configuring a web server.

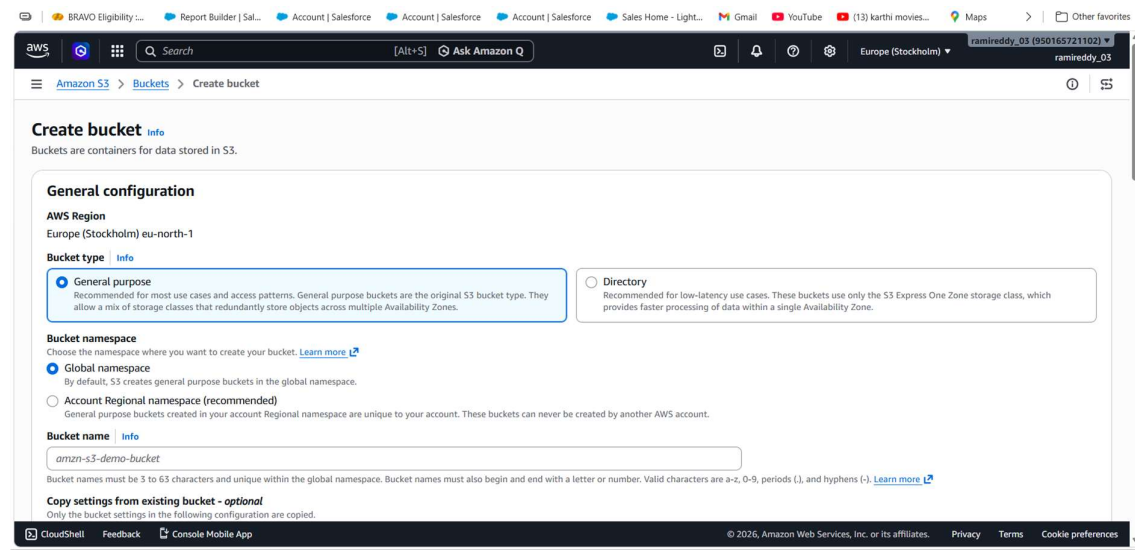
V) Cloud-Based Architecture

The project demonstrates practical cloud computing concepts using AWS services.

Step 1: Creating an S3 Bucket

The first step is to create an Amazon S3 bucket that will contain all website files and resources.

Screenshot 1: Creating a New S3 Bucket



This screenshot shows the bucket creation page in the AWS Management Console. The bucket name selected for this project is:

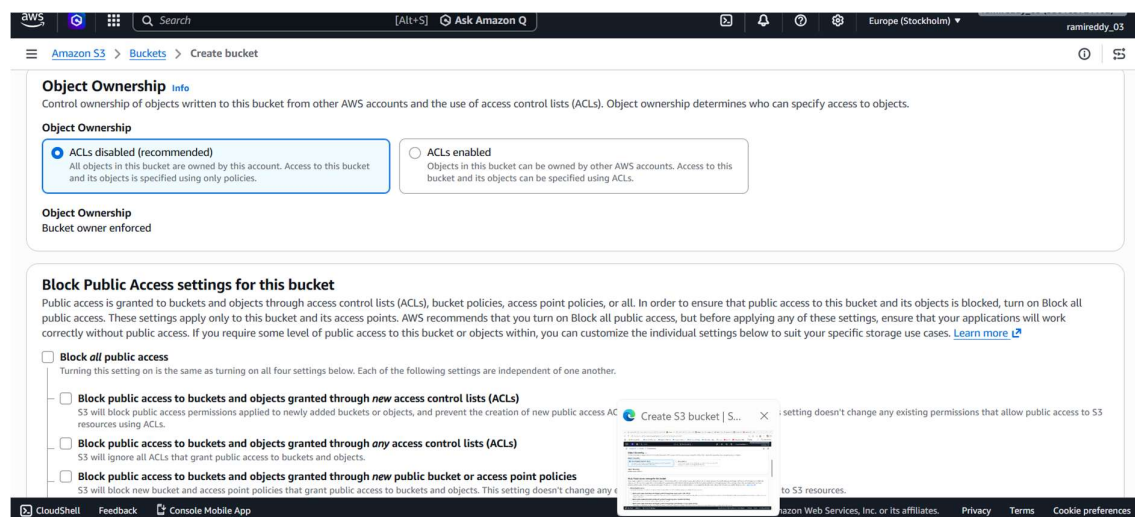
rami-aws-static-website

The selected AWS region is:

Europe (Stockholm) - eu-north-1

Bucket names must be globally unique across AWS.

Screenshot 2: Configuring Object Ownership



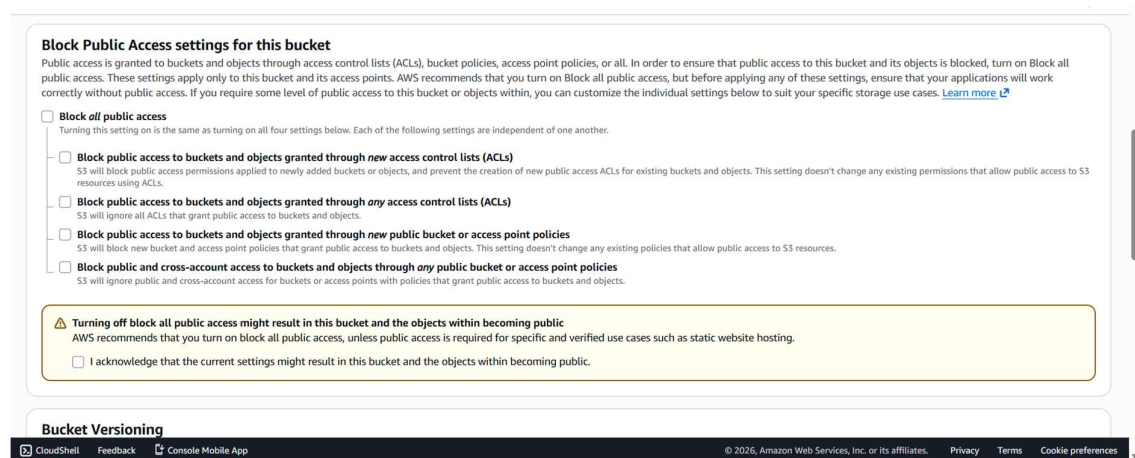
This screenshot shows the Object Ownership configuration during bucket creation.

The selected option is:

ACLs Disabled (Bucket Owner Enforced)

This setting is recommended by AWS because bucket policies provide a more secure and manageable approach to access control.

Screenshot 3: Configuring Public Access Settings



This screenshot displays the "Block Public Access" configuration.

The option:

Block all public access

was disabled to allow website visitors to access the hosted content through the internet.

This step is necessary because a static website must be publicly accessible.

Step 2: Uploading Website Files

After creating the bucket, the website files were uploaded.

Screenshot 4: Uploading Files to the Bucket

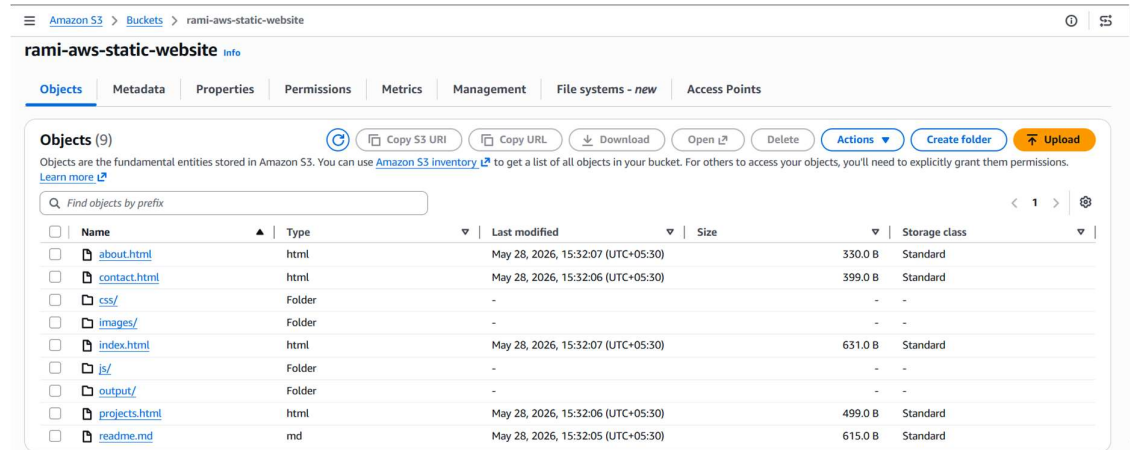
The following files and folders were uploaded:

- index.html
- about.html
- projects.html
- contact.html

- css/
- js/
- images/

These files collectively form the static portfolio website.

Screenshot 5: Uploaded Objects in S3 Bucket



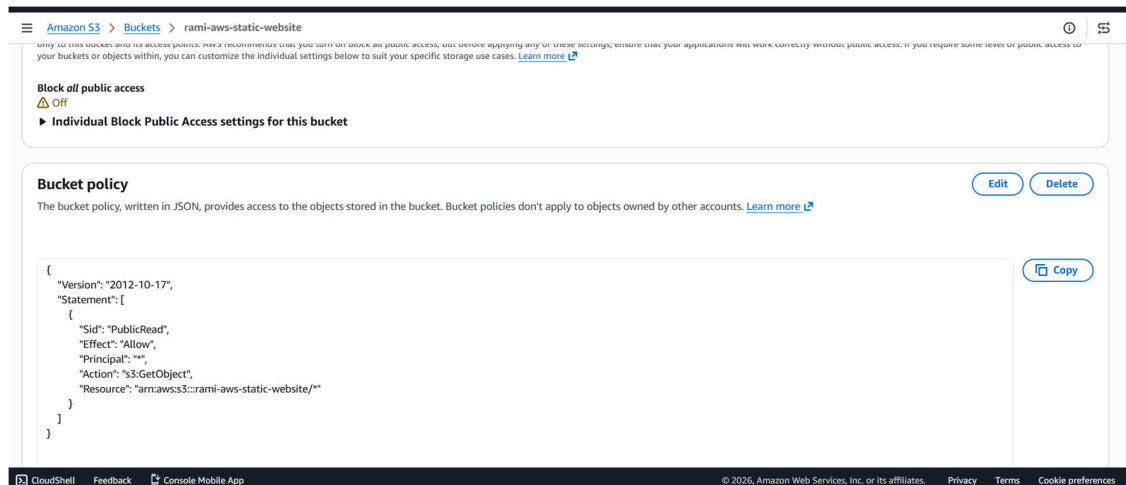
This screenshot shows the successful upload of all website files and folders inside the S3 bucket.

The root directory contains the index.html file, which acts as the default landing page of the website.

Step 3: Configuring Bucket Permissions

To allow public access to website files, a bucket policy was configured.

Screenshot 6: Bucket Policy Configuration



The following policy was added:

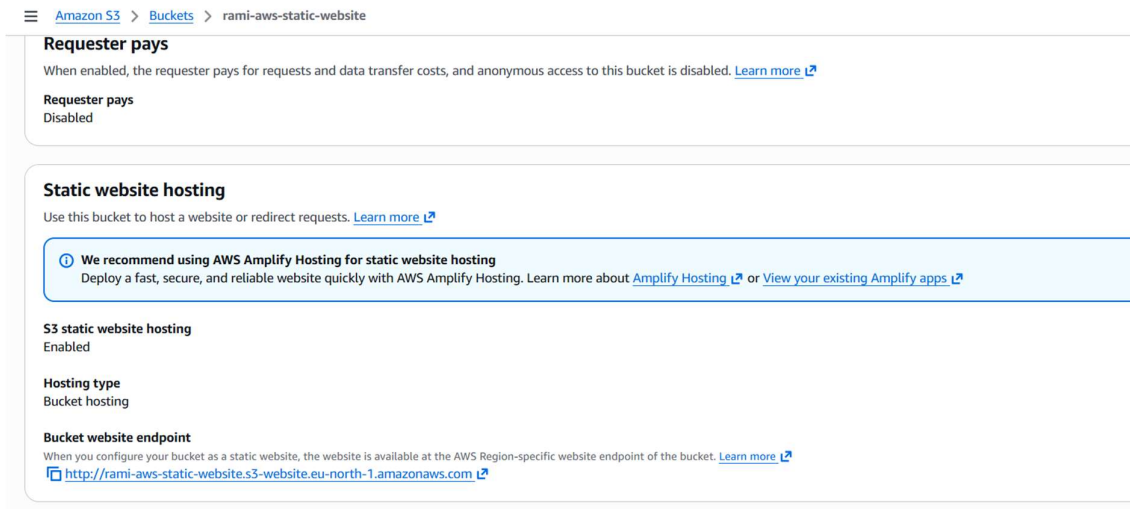
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicRead",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3::rami-aws-static-website/"
    }
  ]
}
```

This policy grants read-only access to website files for all internet users.

Step 4: Enabling Static Website Hosting

AWS S3 provides a dedicated feature for hosting static websites.

Screenshot 7: Enabling Static Website Hosting



The Static Website Hosting option was enabled from the bucket properties page.

Configuration:

Hosting Type:

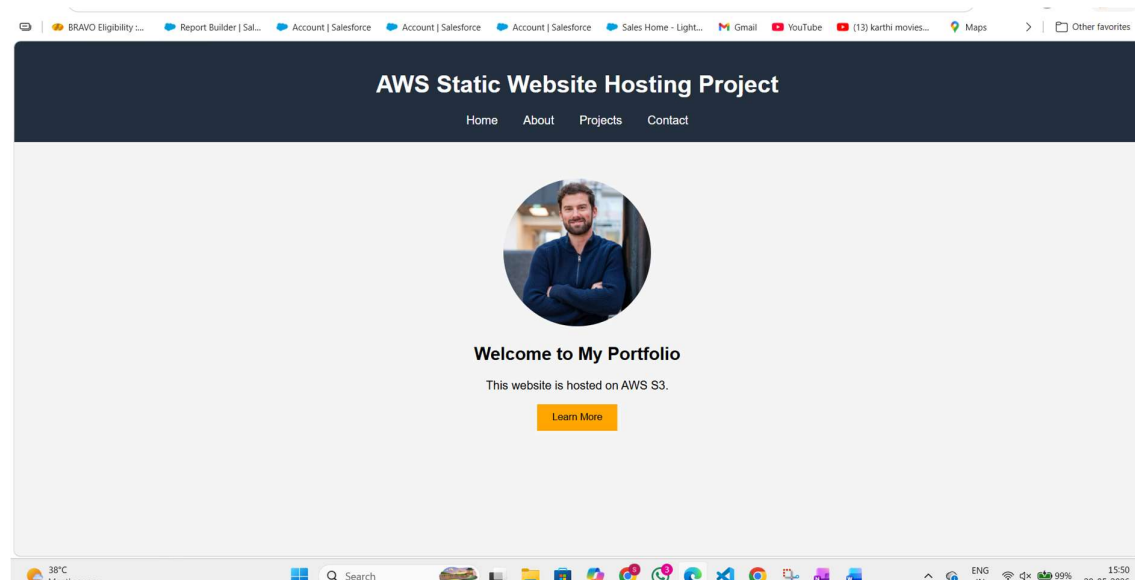
Host a Static Website

Index Document:

index.html

This tells AWS which page should be displayed when users access the website URL.

Screenshot 8: Static Website Hosting Successfully Enabled



This screenshot shows the successful activation of static website hosting.

AWS generated the following public website endpoint:

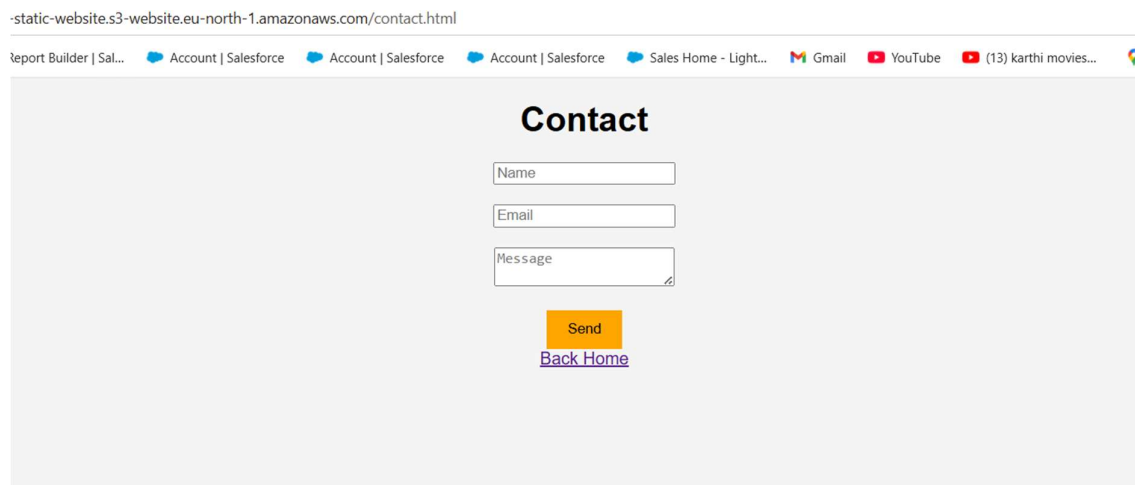
<http://rami-aws-static-website.s3-website.eu-north-1.amazonaws.com>

Users can access the website directly using this URL.

Step 5: Contact Form Integration

The website contains a Contact page that allows visitors to submit messages.

Screenshot 9: Contact Form Page



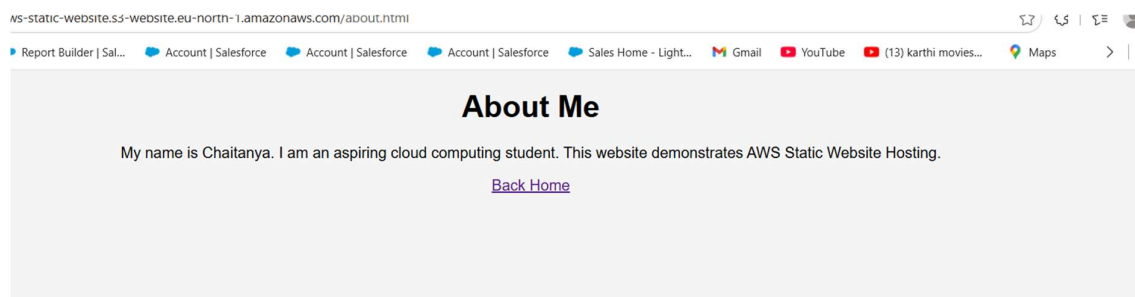
The screenshot shows a web browser window with the URL `static-website.s3-website.eu-north-1.amazonaws.com/contact.html`. The browser's address bar and tabs are visible at the top. The main content area has a light gray background and is titled "Contact" in a large, bold, black font. Below the title, there are three input fields: "Name", "Email", and "Message". The "Message" field is a text area. Below these fields is an orange "Send" button and a purple "Back Home" link.

The contact form collects:

- Name
- Email Address
- Message

Website Output

Screenshot 11: About Page

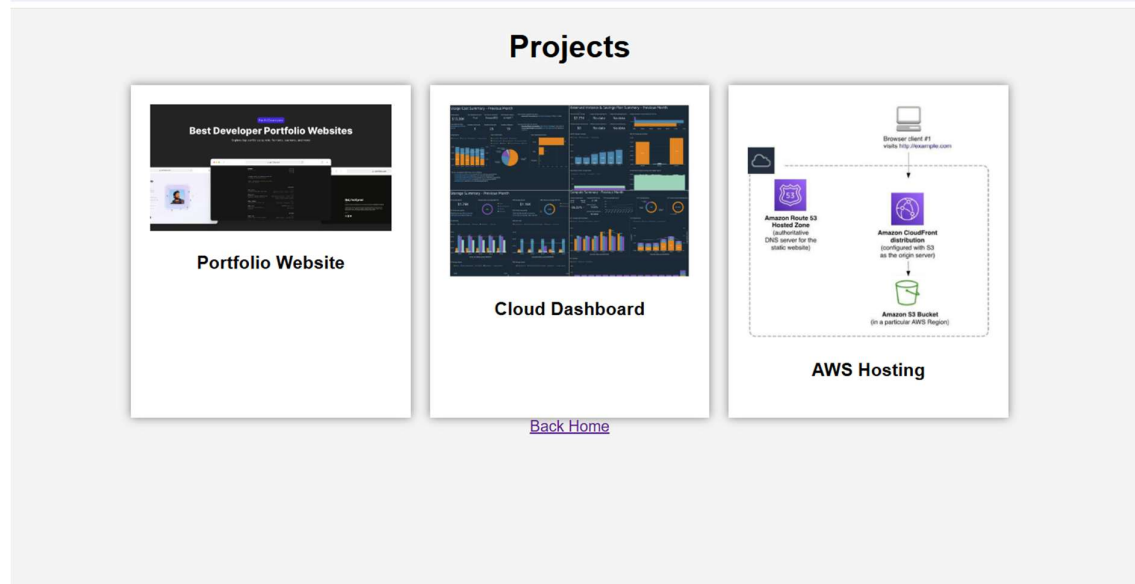


The screenshot shows a web browser window with the URL `static-website.s3-website.eu-north-1.amazonaws.com/about.html`. The browser's address bar and tabs are visible at the top. The main content area has a light gray background and is titled "About Me" in a large, bold, black font. Below the title, there is a paragraph of text: "My name is Chaitanya. I am an aspiring cloud computing student. This website demonstrates AWS Static Website Hosting." Below the paragraph is a purple "Back Home" link.

Provides information about the project and developer.

Screenshot 12: Projects Page

C:/Users/CHITANYA/OneDrive/Desktop/static%20website/projects.html



Displays project cards and portfolio content.

Advantages of AWS S3 Static Website Hosting

- Low hosting cost
- High availability
- Fast deployment
- Scalability
- No server management required
- Secure cloud infrastructure
- Easy maintenance

Learning Outcomes

Through this project, the following concepts were learned:

- Cloud Computing Fundamentals
- AWS S3 Bucket Management

- Static Website Hosting
- Bucket Policies and Permissions
- Public Access Configuration
- Frontend Web Development
- Google Sheets Integration
- Cloud Deployment Techniques

Conclusion

In this project, a fully functional static portfolio website was successfully hosted using Amazon S3. The website was made publicly accessible by configuring bucket permissions and enabling static website hosting. The project also integrated a contact form with Google Sheets for storing visitor messages. This implementation demonstrates the practical use of AWS cloud services for deploying scalable, cost-effective, and highly available web applications.